

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q.1 Solve Any Two of the following.		
A) State and explain any six applications of python.	Level 6	6
B) State which of the following python statements are valid and invalid. a. x = input("enter your name") b. x = input() c. x = input("") d. x = input(" ' ") e. x = input(2) f. x= input("Python" + 53)	Level 3	6
C) What are identity operators, explain with suitable examples.	Level 6	6
Q.2 Solve Any Two of the following.		
A) Differentiate between List, Tuple & Set with suitable examples of each.	Level 6	6
B) Explain any six assignment operators of python with suitable examples of each.	Level 3	6
C) If a four-digit number is input through the keyboard, write a program (using for loop) to display sum of all digits.	Level 4	6
Q.3 Solve Any Two of the following.		
A) If the ages of Raj, Ketan and Kunal are input through the keyboard, write a program to determine the a) Youngest of the three. b) Oldest of three	Level 6	6
B) Explain count() methods of list. Write a python program which will take name and subject as input from students (from class of 75 students) and at the end it should display names of all students and total number of students who opted for subject Python Programming.	Level 4,5	6
C) Explain for loop with suitable example?	Level 3,4	6

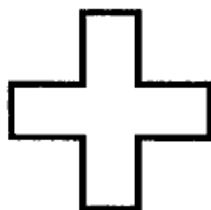
Q.4 Solve Any Two of the following.

- A) Explain concept of OOP in Python with suitable example. Also Explain concept of inheritance in detail. **Level 5,6** **6**
- B) Write a python program to print following pattern (using loop statement) **Level 3** **6**
- ```

 * * *
 * *
 *
```
- C) If any integer is input through the keyboard. Write a python program to find out whether it is a prime number or non-prime number. **Level 4** **6**

**Q.5 Solve Any Two of the following.**

- A) What is turtle in Python? Write a program to draw following pattern using same. **Level 4,5** **6**



- B) A program has to be written in python which will take two numbers as input from user and perform the division in such a way that program should not terminate (because of ZeroDivisionError) even if user input is zero. **Level 4,5** **6**
- C) If any two numbers are the input by user, write a program to design a simple arithmetic calculator (addition, subtraction, multiplication and division) with the help of OOP concepts. **Level 4** **6**

**\*\*\* End \*\*\***